

# ComicCraft AI: AI-Powered Five-Panel Comic Creator

**Project Name:** ComicCraft AI

**Team ID:** 6ab3ed02b5170a9b58b3aa3a

**Date:** 02 October 2026

## Project Description

ComicCraft AI is a web-based application that uses generative AI to create personalized five-panel comic stories and illustrations from user prompts. The application uses Python, FastAPI, Google Gemini AI models, Pollinations image generation, HTML/CSS/JavaScript, Jinja2, Pillow, comic lettering, and PDF export.

## Example Scenarios

- A user enters a story idea about a college student exploring an abandoned railway station. Gemini develops the story into five stages and Pollinations generates the corresponding comic panels.
- A user selects a mysterious and adventurous tone with a polished 2D cartoon comic style. The system uses these inputs while generating the story and visual scenes.
- A user previews the completed five-panel comic with narration and dialogue and can export the final comic as a PDF.

## **Milestone 1: Model Selection and Architecture**

- Understand the project requirements for structured five-panel story generation, image generation, narration/dialogue, preview, and PDF export.
- Use Gemini for structured story and comic-script generation.
- Use Pollinations for AI comic-panel image generation.
- Define the architecture connecting the browser frontend, FastAPI backend, AI services, image processing, lettering, layout, and PDF export.

## **Technical Prerequisites**

- Python
- FastAPI and Uvicorn
- HTML, CSS, JavaScript and Jinja2
- Google Gemini API
- Pollinations API
- Pillow
- PDF export libraries
- Git and GitHub

## **Milestone 2: Core Functionalities**

- Story generation from the user's story idea and character/setting details.
- Exactly five comic panels arranged as Beginning, Development, Turning Point, Climax, and Ending.
- Character-consistency instructions carried across the generated scenes.
- Narration and dialogue generation and comic lettering.
- Comic preview and PDF export.

## **Milestone 3: Main Application Development**

- FastAPI routes receive the user inputs and coordinate the comic-generation workflow.
- Gemini modules generate the structured story and comic narration.
- The image-generation module creates panel images through the configured Pollinations backend.
- Layout and exporter modules assemble the final comic output.

## Milestone 4: Frontend Development

The frontend provides fields for Story Idea / Prompt, Main Character, Setting / Location, Tone, and Art Style. Jinja2 templates are used for the input page, comic preview, and export-success flow.

The result displays the five generated panels in story order, with narration/dialogue added to make the output readable as a complete comic.

## Milestone 5: Local Setup and Deployment

### Local setup used for the project:

```
C:\Users\acer\OneDrive\Documents\comicCraft>python -m venv .venv
C:\Users\acer\OneDrive\Documents\comicCraft>.venv\Scripts\activate
(.venv)>pip install -r requirements.txt
```

Create the local .env from .env.example and keep real API keys private. The public repository contains .env.example rather than the private .env file.

### Render deployment:

**Hosting Provider:** Render

**Build Command:** pip install -r requirements.txt

**Start Command:** uvicorn app.main:app --host 0.0.0.0 --port \$PORT

**Hosted URL:** <https://comiccraft-r3m7.onrender.com>

**Login Required:** No login required

## Deployment and Repository Details

**GitHub Repository:** <https://github.com/devilanand73055-dotcom/ComicCraft>

**Demo Video:** [https://drive.google.com/file/d/1-ONYxu8H-\\_ndzGAzGqhSSbf5owSrPanW/view?usp=sharing](https://drive.google.com/file/d/1-ONYxu8H-_ndzGAzGqhSSbf5owSrPanW/view?usp=sharing)

**Application:** ComicCraft AI converts a simple story idea into a complete five-panel visual comic.

## Project Structure

```
ComicCraft/  
├── app/  
│   ├── main.py  
│   ├── routes.py  
│   ├── models.py  
│   ├── config.py  
│   ├── gemini_flash.py  
│   ├── gemini_pro.py  
│   ├── image_generator.py  
│   ├── layout_builder.py  
│   ├── lettering.py  
│   ├── exporters.py  
│   └── comic_script.py  
├── templates/  
├── requirements.txt  
├── .gitignore  
└── .env.example
```

## Website Pages, Inputs and Results

**Home / Comic Generator:** The main interface accepts the story prompt, character, setting, tone, and art style and provides the Generate Comic action.

**Comic Preview:** Displays the generated five-panel sequence with narration and dialogue.

**Export:** The completed comic can be assembled and exported as a PDF.

**Result Sequence:** Beginning → Development → Turning Point → Climax → Ending.

## Conclusion

ComicCraft AI is a generative AI web application that transforms a simple story idea into a structured five-panel comic. Gemini is used for story generation, Pollinations for comic image generation, and FastAPI coordinates the application workflow. The project also includes character-consistency prompting, narration and dialogue, comic lettering, preview, and PDF export.

The application combines Python, FastAPI, HTML, CSS, JavaScript, Jinja2, Pillow, AI services, GitHub, and Render into a complete workflow. It is designed to make comic creation more accessible to students, beginner comic creators, and users who want to convert their ideas into structured visual comics.